

SO project - Stochastic Machine Replacement Problem

Yiping Fang, LGI, CentraleSupélec

April 4, 2025

Problem statement

At the end of each production cycle (e.g., seasonal), a cheese production line must decide whether to continue using a machine or replace it with a new one. A machine in the production line at cycle t has a corresponding efficiency state $s_t \in S = \{1, 2, \dots, 10\}$ depending on the degradation level of the machine. A higher value of s_t represents a more degraded state and thus low productivity of the production line.

We know the machine's state at the first cycle $s_1 = 1$, and it has probability $p = 0.9$ to go to efficiency state $s_{t+1} = \min\{s_t + 1, 10\}$ and probability $1 - p$ to go to efficiency state $s_{t+1} = \min\{s_t + 2, 10\}$ if not replaced by a new one.

At each efficiency state $s \in S$, it produces $y(s) = 8 + s - 0.15s^2$ tons of cheese over the corresponding production cycle. We assume the machine must be replaced upon completion of the production cycle $s_t = 10$ since it becomes too degraded to be productive. The net cost of replacing a machine is $c = 500$ k€ and the profit contribution of cheese is $m = 150$ k€ per ton.

This is an infinite-horizon stochastic model with time t measured in production cycles. The state is.

$$\begin{aligned} s_t &= \text{production efficiency state of a machine at production cycle } t \\ s_t &\in \{1, 2, \dots, 10\} \end{aligned}$$

The action is

a = replacement decision “keep” or “replace”

$$a \in A(s_t) = \begin{cases} \{\text{keep}, \text{replace}\} & s_t < 10 \\ \{\text{replace}\} & s_t = 10 \end{cases}$$

The state transition function is

$$T(s_t, a, s_{t+1}) = \begin{cases} p & a = \text{keep}, s_{t+1} = s_t + 1, s_t \leq 8 \\ 1 - p & a = \text{keep}, s_{t+1} = s_t + 2, s_t \leq 8 \\ 1 & a = \text{keep}, s_{t+1} = 10, s_t = 9 \\ 1 & a = \text{replace}, s_{t+1} = 1, \forall s_t \end{cases}$$

The reward function is

$$r(s, a) = \begin{cases} m \cdot y(s) & a = \text{keep} \\ m \cdot y(s) - c & a = \text{replace} \end{cases}$$

where

$$y(s) = 8 + s - 0.15s^2$$

Question:

- Write down the Bellman optimality equation of the value function.
- What replacement policy maximizes the expected long-term cumulative profit? (Solve the problem using value iteration, policy iteration, or linear programming.)
- Test the sensitivity of optimal policies to the state transition probabilities p .